Experiment-16

Aim

To perform edge detection on an image using the Canny Edge Detection algorithm in Python with OpenCV.

Procedure

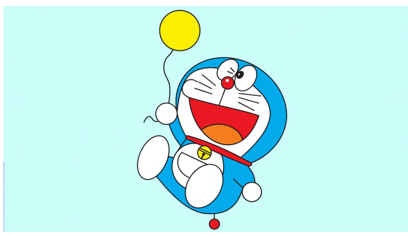
1. Import the OpenCV library.
2. Read the input image.
3. Convert the image into grayscale.
4. Apply Gaussian Blur to reduce noise.
5. Apply the Canny edge detector by specifying lower and upper threshold values.
6. Display the original image and the detected edges.
7. Save the output image if required.

Code:

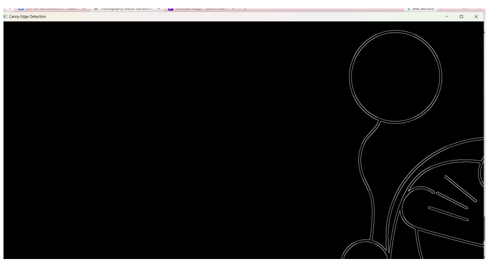
```
import cv2

image = cv2.imread("sample.jpg")

if image is None:

    print("Error: Image not found!")

    exit()

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

blur = cv2.GaussianBlur(gray, (5, 5), 0)

edges = cv2.Canny(blur, 100, 200)

cv2.imshow("Original Image", image)
```

```
cv2.imshow("Canny Edge Detection", edges)

cv2.waitKey(0)

cv2.destroyAllWindows()
```

Sample Input:



Output:



Results:

The Canny edge detection algorithm was successfully applied to the input image. The edges were accurately detected after grayscale conversion and Gaussian smoothing, producing a clear edge map of the image.